

Virgo SLM V1.0

Building a 120M Parameter Transformer Language Model from Scratch

A Self-Learning Engineering Project

Tagline: *"Not Everybody Needs The Blueprint."*

Project Vision

Write the report as the engineering diary of a first-year IIT Dharwad student. Focus on what was learned, what failed, and how each engineering challenge was solved. Avoid marketing language or exaggerated claims. The tone should be honest, curious, and realistic.

Three Development Phases

■ Phase I – Language Learning

~2 Billion pretraining tokens. Build Virgo Base by collecting datasets, cleaning text, training a custom 45K Byte-Level BPE tokenizer, implementing a decoder-only Transformer, and learning grammar, vocabulary, syntax, semantics, and next-token prediction.

■ Phase II – Chat Style Tuning

~230 Million conversational tokens. Transform Virgo Base into Virgo Chat using conversational datasets. Focus on dialogue quality, assistant identity, writing quality, context handling, and natural conversations.

■ Phase III – Instruction & Response Optimization

~278 Million instruction-tuning tokens. Explain that a large custom instruction dataset was manually designed and curated by the author over many months. Highlight the engineering effort spent creating prompt–response datasets, cleaning data, balancing tasks, and improving instruction following, structured outputs, writing, coding, summarization, information extraction, and response consistency. Output: Virgo Instruct (IFT).

Recommended Chapters

- 1. Why I Started Virgo
- 2. My Learning Journey
- 3. Project Roadmap
- 4. Dataset Engineering
- 5. Tokenizer
- 6. Transformer Architecture
- 7. Training Journey
- 8. Challenges & Failed Experiments
- 9. Demonstration Examples
- 10. Capability Assessment
- 11. Current Limitations

- 12. Future Vision
- 13. Personal Message

Capability Assessment (Suggested Table)

Capability	Status	Comments
Text Generation	■■■■■ Strong	Produces coherent paragraphs and natural language.
Language Coherence	■■■■■ Strong	Maintains readable and consistent responses.
Basic Conversation	■■■■■ Strong	Good for everyday interactions.
Writing & Summaries	■■■■■ Strong	Strongest area of the current model.
General Knowledge	■■■■■ Moderate	Reasonable coverage but may contain factual mistakes.
Instruction Following	■■■■■ Improving	Noticeably improved after Phase III but complex instructions remain difficult
Reasoning	■■■■■ Developing	One of the biggest areas for future improvement.
Programming	■■■■■ Developing	Limited to simple code generation; struggles with debugging and advanced
Mathematics	■■■■■ Limited	Not reliable for advanced mathematical reasoning.
Long Context	■■■■■ Limited	Context window limits long conversations.
Memory	■	No persistent memory.
Web Search	■	Not supported.
Image Generation	■	Not supported.
File / Tool Use	■	Not supported.

Current Limitations

Clearly mention that Virgo currently does not support web search, image generation, persistent memory, file creation, tool use, long-context conversations, continuous multi-session conversations, advanced reasoning, or reliable advanced mathematics. Explain that the current interaction style is prompt → response.

Future Vision

Describe future goals: stronger reasoning, improved instruction following, larger datasets, larger context windows, multimodal capabilities, retrieval augmentation, tool use, memory, and eventually scaling Virgo beyond the Small Language Model category while keeping it efficient.

Closing Message

End with a heartfelt note reconnecting to the project tagline: **"Not Everybody Needs The Blueprint."** Explain that Virgo was built independently as a self-learning journey. Mention the failed training runs, CUDA errors, preprocessing bugs, debugging sessions, and the patience required. Encourage other students to start building, experimenting, and learning even without having the perfect roadmap.